

YouTube Trending Video Analysis

Predicting Viral Content Using NLP, Clustering & XGBoost

Suparna Deshmukh · US Region Dataset · 40,949 Records · 2024

Overall Accuracy

87.86%

XGBoost classifier

AUC Score

0.8725

ROC curve area

Viral F1 Score

0.93

Precision-Recall balance

Clusters Found

4

K-Means (k=4)

1. Project Overview

This project delivers a complete end-to-end machine learning pipeline on the YouTube Trending Video Dataset (US region, sourced from Kaggle). The central objective is to predict whether a video will qualify as **viral** — defined as appearing on the YouTube trending page for three or more consecutive days — using a combination of natural language processing, unsupervised clustering, and gradient-boosted classification.

The pipeline spans six analytical stages: exploratory data analysis, SQL-based aggregation, NLP feature engineering, K-Means clustering, XGBoost classification, and SHAP-based model explainability. All stages feed sequentially into a final trained model and interpretable feature importance report.

Dataset Summary

Property	Detail
Source	Kaggle — YouTube Trending Video Dataset (datasnaek/youtube-new)
Region	United States (USvideos.csv)
Records	40,949 rows × 16 columns
Coverage	Daily trending snapshots across multiple months
Key Fields	title, tags, views, likes, dislikes, comment_count, category_id, trending_date

Methodology Pipeline

Stage	Technique	Output
1. Data Exploration	Pandas, Matplotlib, Seaborn	4 EDA charts
2. SQL Analysis	SQLite + Pandas	3 aggregated result tables
3. Target Engineering	Trending day count threshold	is_viral binary column
4. NLP Engineering	Text cleaning + TF-IDF (100 feat.) + 11 numeric features	Feature matrix X (111 cols)
5. Clustering	K-Means, elbow method (k=4)	4 named video archetypes
6. Classification	XGBoost + SHAP explainability	Trained model + importance scores

2. Exploratory Data Analysis & SQL Findings

• **Category Distribution:**

Entertainment, Music, and Comedy dominate the US trending page by volume. These three categories collectively account for over 60% of all trending video appearances.

• **View Count Distribution:**

Heavily right-skewed — the vast majority of trending videos receive between 500K and 5M views, while a small elite exceeds 50M. A log scale is necessary to visualise the spread meaningfully.

• **Like-to-View Ratio:**

The median like-to-view ratio across all trending videos is approximately 4–6%, meaning roughly 1 in 20 viewers actively clicked Like. Gaming and Comedy categories show the highest ratios, indicating more passionate and engaged audiences.

• **Day-of-Week Patterns:**

Trending video counts are distributed fairly evenly across weekdays, with no single day showing a dominant spike — suggesting YouTube's trending algorithm refreshes continuously rather than following a publishing-day pattern.

SQL Analysis — Key Query Results

SQL Query	Top Finding
Top 10 categories by avg views	Music leads with highest average views per video by a significant margin
Avg engagement rate per category	Gaming and Comedy have highest engagement rates despite lower raw view counts
Top 20 channels by trending days	Dominant channels trend through volume of videos, not single breakout hits

3. Feature Engineering & K-Means Clustering

Target Variable — `is_viral`

A binary target column was engineered by counting how many days each unique `video_id` appeared in the trending dataset. Videos with **3 or more trending days** were labelled viral (1); all others were labelled non-viral (0). This threshold produced a heavily imbalanced dataset: 96% viral, 4% non-viral — a challenge addressed during model training via the `scale_pos_weight` parameter in XGBoost.

Feature Matrix — X (111 Total Features)

Feature Group	Count	Description
TF-IDF (title + tags)	100	Top 100 word/phrase scores from cleaned title and tags text
Like-to-view ratio	1	Likes divided by views — measures engagement quality
Dislike-to-view ratio	1	Dislikes divided by views — captures controversy signal
Comment-to-view ratio	1	Comments divided by views — measures discussion depth
Like-to-dislike ratio	1	Overall audience sentiment balance
Title length	1	Character count of cleaned video title
Title word count	1	Number of words in the cleaned title
Tag count	1	Number of tags creator added to the video
Log views / likes / comments	3	Log-scaled raw engagement counts
Category ID	1	Numeric YouTube content category identifier

K-Means Clustering Results (k=4, Elbow Method)

The elbow method tested k=2 through k=10. Inertia dropped sharply from k=2 (105,934) to k=4 (56,792) then flattened significantly, identifying **k=4** as the optimal number of clusters. Four distinct video archetypes emerged:

Cluster	Avg Views	Videos	Avg Like Ratio	Viral Rate	Interpretation
Engaged Community	1.45M	10,778	0.068	97.4%	Small but highly loyal audience
Standard Trending	1.65M	29,289	0.022	95.0%	Bulk of typical trending content
Mainstream Hits	31.2M	808	0.045	99.4%	Genuinely viral popular videos
Viral Giants	99.2M	74	0.046	100.0%	Mega-viral once-in-a-generation

The Engaged Community cluster is notable for having the **highest like-to-view ratio (0.068)** despite the lowest average views — suggesting niche but deeply engaged audiences. Viral Giants (74 videos, 100% viral rate, 99M avg views) represent an extreme outlier class that the model treats as a near-certain viral signal.

4. XGBoost Classification & Model Performance

Model Configuration

Parameter	Value	Rationale
n_estimators	300	300 trees — strong learning without excessive runtime
max_depth	6	Deep enough to capture interactions, avoids overfitting
learning_rate	0.1	Standard conservative learning step size
subsample	0.8	80% of rows per tree — adds regularisation
colsample_bytree	0.8	80% of features per tree — reduces correlation between trees
scale_pos_weight	~0.04 (auto)	Compensates for 96%/4% class imbalance
Train/Test Split	80% / 20%	Stratified split preserving class proportions

Classification Report

Class	Precision	Recall	F1 Score	Support	Interpretation
Not Viral (0)	0.22	0.72	0.34	351	Weak — few not-viral examples in training data
Viral (1)	0.99	0.89	0.93	7,839	Strong — model reliably identifies viral content
Weighted Avg	0.95	0.88	0.91	8,190	Inflated by dominant viral class

Overall Accuracy	AUC Score	Viral F1	Not Viral F1
87.86%	0.8725	0.93	0.34

ROC Curve & AUC Interpretation

The AUC score of **0.8725** is the most honest performance metric here because it measures the model's ability to rank viral above non-viral videos independently of class imbalance. A score of 0.87 confirms the model has learned genuinely useful patterns — well above the 0.50 random baseline — though meaningful improvement toward

0.90+ is achievable with the fixes outlined in Section 5.

5. SHAP Explainability & Conclusions

Top 10 Features by SHAP Importance

Rank	Feature	SHAP Score	Meaning
#1	audio	0.778	Music videos with 'Official Audio' in title — dominant viral pattern
#2	ft	0.745	Featuring collaborations — music artist collabs trend strongly
#3	youtube	0.524	YouTube-specific content (Rewind, creator collabs)
#4	science	0.511	Science channels (Vsauce etc.) trend with loyal audiences
#5	interview	0.420	Celebrity/political interviews spike around news cycles
#6	hair	0.391	Beauty and lifestyle content trends consistently
#7	tv	0.360	TV show clips and recaps drive consistent trending
#8	home	0.355	Home improvement / lifestyle category signal
#9	news	0.342	Breaking news and commentary content
#10	movies	0.334	Trailers and movie-related content

Key SHAP Insight: All top 10 features are TF-IDF text tokens — none of the numeric engagement features (log_views, like_ratio etc.) appear in the top 10. This reveals the model predicts virality primarily from **content type** rather than engagement metrics. Music collaborations, science content, and interview videos are so consistently viral in this dataset that content category alone is a stronger signal than view counts or ratios.

Key Conclusions

- **Content type outperforms engagement metrics.**

The model's strongest predictors are words in titles and tags, not view counts or like ratios. Knowing a video is a music collaboration or science video is more predictive of virality than how many views it has accumulated.

- **Music dominates US trending.**

The words 'audio' and 'ft' (featuring) are the two most important features, confirming that music video releases — particularly collaborations — drive a disproportionate share of US trending appearances.

- **Class imbalance is the core modelling challenge.**

With 96% of videos labelled viral, the model's 87.86% accuracy is partly superficial. The true challenge — and biggest opportunity for improvement — is distinguishing the 4% of non-viral videos more reliably (current F1: 0.34).

- **Four audience archetypes exist in the data.**

K-Means clustering reveals videos naturally separate into Engaged Community, Standard Trending, Mainstream Hits, and Viral Giants — with dramatically different view counts and engagement profiles across the four groups.

Recommended Next Steps

Limitation	Fix
96% viral — easy to predict	Raise is_viral threshold to 7+ trending days
Model relies on content type	Retrain using numeric features only

US-only dataset	Merge all country CSVs for global patterns
No temporal features	Add publish hour and days-to-trending as features
Class imbalance persists	Apply SMOTE oversampling on training set

Final Takeaway: A YouTube video is most likely to go viral if it is a music collaboration (featuring another artist), belongs to a high-engagement category such as Science or Entertainment, and accumulates strong early likes relative to its view count. Content type is a stronger predictor of virality than raw engagement numbers in this dataset.